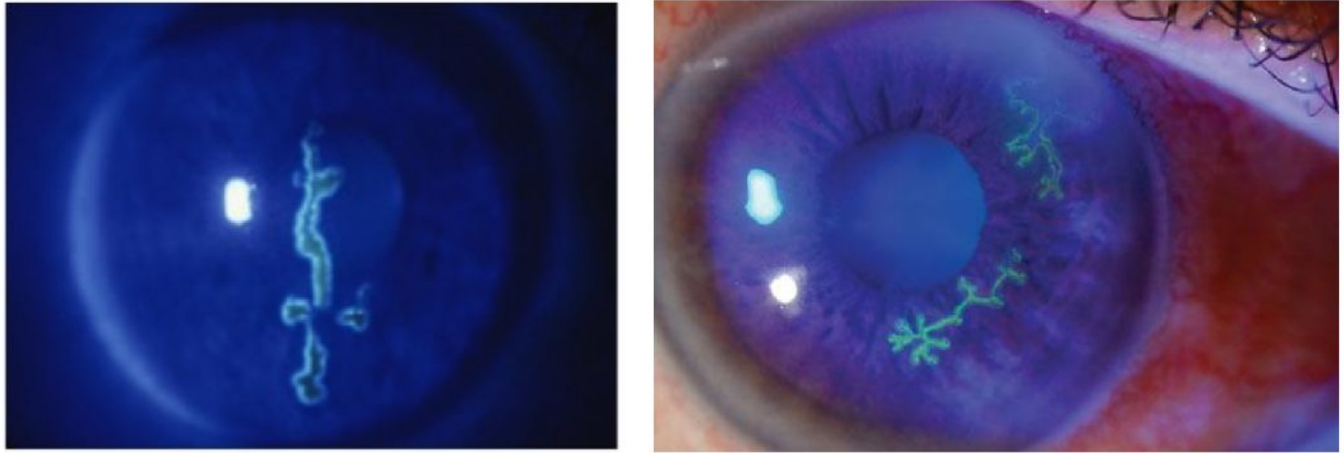


# HSV is not a single-layer process

Epithelial viral replication and deeper inflammation require different treatment reasoning.



Dendritic ulcers (under fluorescein stain + cobalt blue filter)

Branching fluorescein-stained ulcers with terminal bulbs illustrate epithelial HSV disease. Reduced corneal sensation and recurrent unilateral episodes strengthen the clinical diagnosis; the embedded caption is retained from the original Anki image.

## Use the epithelial-stromal distinction

Active epithelial HSV is treated with topical antiviral; steroid alone can increase viral replication and enlarge an epithelial ulcer. Stromal disease may require supervised topical steroids with systemic antiviral cover. Oral acyclovir prophylaxis may be considered for frequent recurrences, immunocompromise or visual-axis scarring. Repeated episodes can affect deeper layers and leave scars, so a familiar history should not replace examination of the current lesion. Long-term topical acyclovir is avoided because of toxicity.

# HZO and fungal clues

Skin distribution, organic trauma and lesion morphology refine the likely cause of keratitis.

## Different nonbacterial pathways

| Clue                           | Interpretation and action                                           |
|--------------------------------|---------------------------------------------------------------------|
| V1 vesicular rash              | HZO; treat shingles with systemic antiviral and assess the eye      |
| Nasal-tip involvement          | Hutchinson sign; nasociliary involvement increases ocular concern   |
| Plant injury; feathery margins | Suspect fungus; obtain corneal scraping and fungal culture          |
| Confirmed Fusarium             | Filamentous fungus; natamycin rather than antibacterial drops alone |

## Read the edges carefully

HZO epithelial lesions tend to be finer with tapered ends and lack HSV terminal bulbs. Fungal ulcers may be slower in onset and have satellite infiltrates, particularly after organic-matter trauma or steroid exposure. Neither a rash nor a trauma history establishes the depth or severity of ocular disease: inspect the cornea and anterior chamber, obtain appropriate samples and tailor therapy.